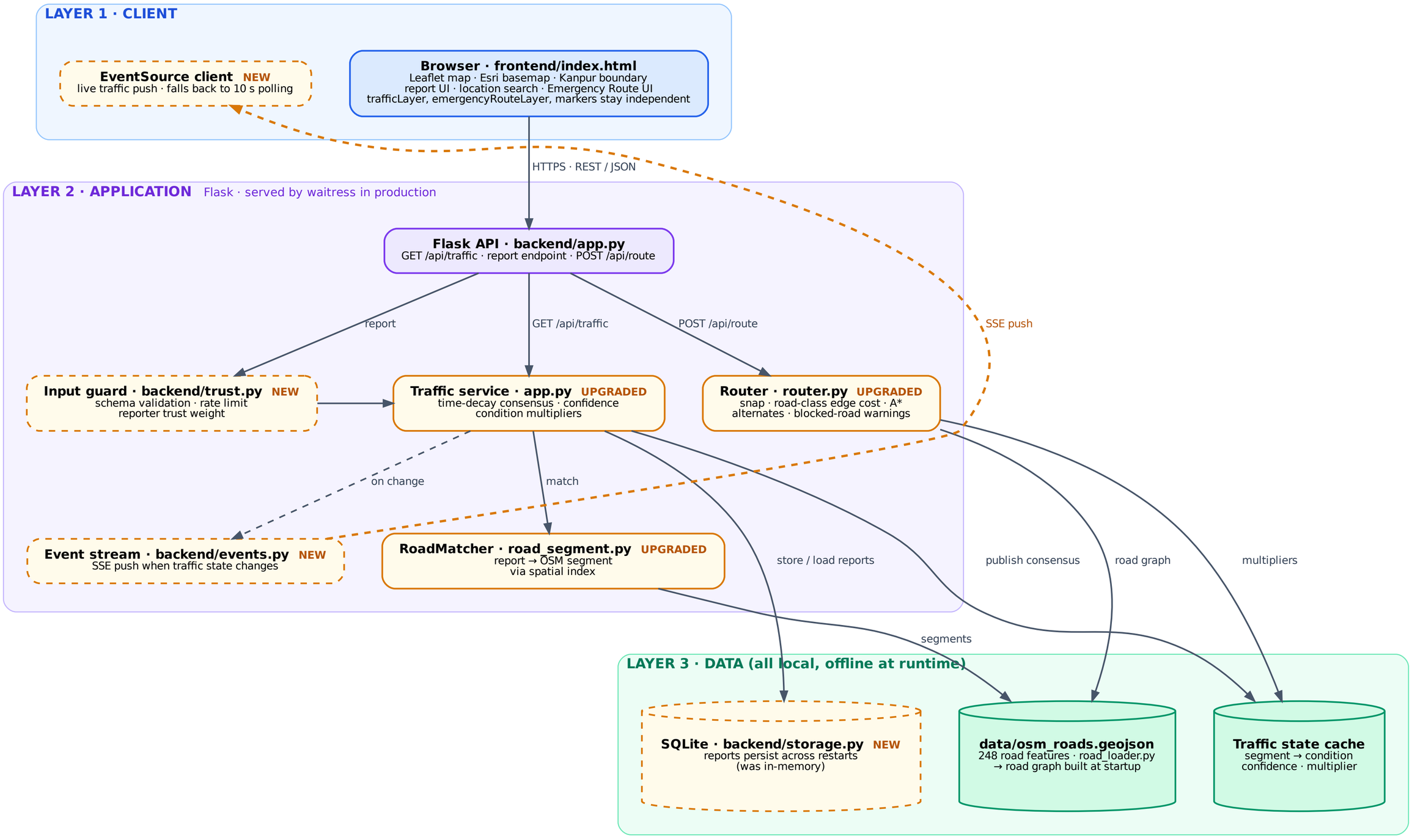
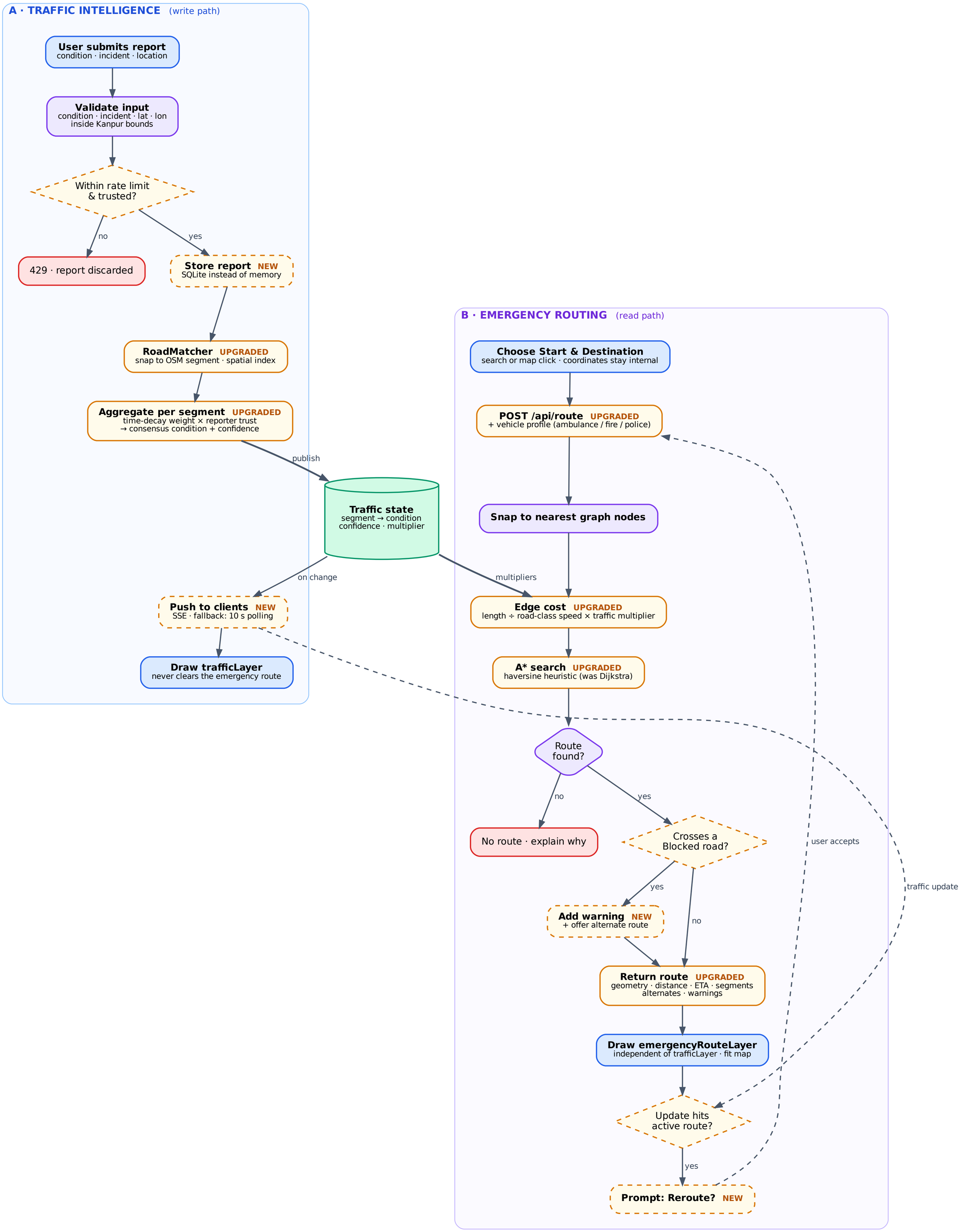
NEXUS · System Architecture (v2)

Community traffic intelligence + emergency routing, Kanpur · **dashed amber = NEW** **solid amber = UPGRADED**



NEXUS · Workflows (v2)

Write path (reports) and read path (routing) share one traffic state · **dashed amber = NEW** **solid amber = UPGRADED**



NEXUS · Improvement Plan (timed)

Estimates for one student developer · run the full test suite (95/95 today) after every step

PHASE 1 · Day 1–2

Make it safe & durable

1 · Persist reports in SQLite
backend/storage.py · 30-min expiry becomes a query filter

2 · Input guard
backend/trust.py · schema validation + rate limit per IP

3 · Blocked-road warning
flag routes crossing Blocked segments instead of silent ×100

4 · Production basics
waitress server · /health · logging · CI runs tests on push

then

PHASE 2 · Day 3–7

Make it fast & realistic

5 · Live push with SSE
backend/events.py · keep 10 s polling as fallback

6 · A* + realistic ETA
cost = length ÷ road-class speed × multiplier

7 · Spatial index in RoadMatcher
STRtree lookup instead of scanning every segment

8 · Time-decay consensus
half-life ≈ 10 min, weighted by reporter trust

then

PHASE 3 · Week 2–3

Make it smarter

9 · Alternates + vehicle profiles
ambulance / fire / police cost tables

10 · Reroute prompt
new report hits the active route → ask to reroute

11 · Confirm / deny on reports
community votes raise or lower confidence

12 · Grow beyond Central Kanpur
export wider OSM area to GeoJSON, rebuild graph